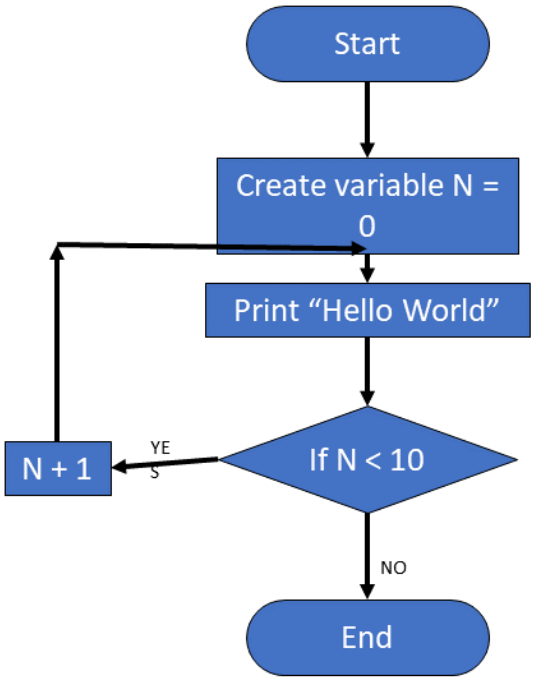
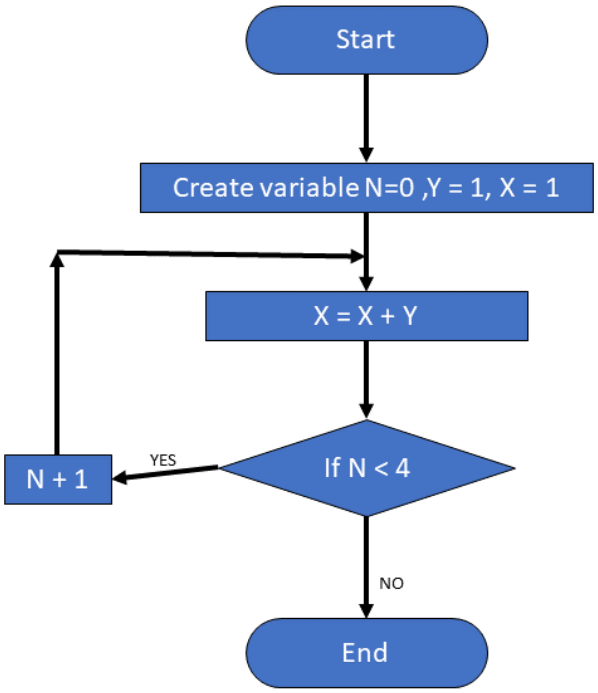
SHEET 1

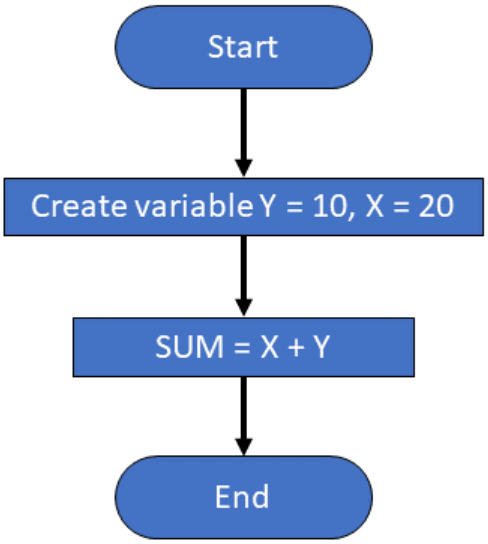
1 -.draw a flowchart Print Hello World 10 times.



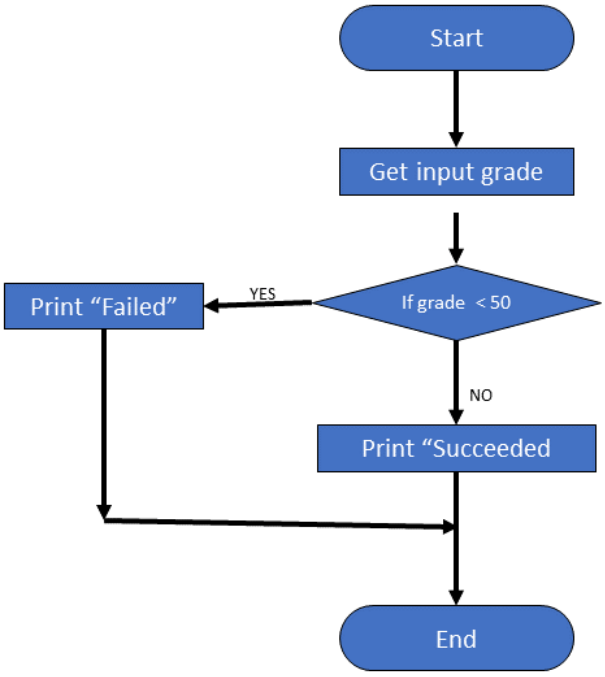
2- . draw a flowchart thats Find the sum of 5 numbers.



3-. draw a flowchart thats Add 10 and 20 and put it in sum.



4- draw a flowchart to determine the state of a student (succeeded of failed) depending on his mark in the exam, full .make is assumed to bo 100.



5- convert these numbers:

- 10 1110 1010
  - H ➔ 2EA
- FFEA
  - B➔ 1111 1111 1110 1010
- 76
  - B ➔ 0100 1100
- 84
  - H ➔ 54

6- The equivalent resistance of resistors connected in series is calculated by adding the resistances of individual resistors. Req = R1 + R2

```
# include <iostream>
using namespace std;

int main(){
    float R1, R2;
    cout << "enter R1 value: ", cin >> R1;
    cout << "enter R2 value: ", cin >> R2;
    cout << "equivalent resistance when connection is: " << endl;
    cout << "in series:   " << R1+R2 << endl;
    cout << "in parallel: " << (R1*R2)/(R1+R2);
    return 0;
}
```

7- Write a C++ program Count Change to count change. Given _
,the number of quarter, dimes

nickels and cents the program should output the total as a
.single value in dollars and cents

```
# include <iostream>
using namespace std;

int main(){
    // define variables
    float quarter, dime, nickel, cent, total;
    // get inputs
    cout << "enter quarters: " ,cin >> quarter;
    cout << "enter dime: " ,cin >> dime;
    cout << "enter nickel: " ,cin >> nickel;
    cout << "enter cent: " ,cin >> cent;
    // calculate total in dollars and cents
    total = ((dime * 10) + (nickel * 5) + (quarter * 25) +
cent)/100;
    cout << "total in dollars and cents: " << total;
    return 0;
}
```

8 -The Pythagorean Theorem states that the sum of the squares
of the two sides of the right angle triangle is equal to the square
.of the hypotenuse

```
# include <iostream>
# include <cmath>
using namespace std;

int main(){
    float m, n, a, b, c;
    cout << "enter m value: " , cin >> m;
    cout << "enter n value: " , cin >> n;
    a = pow(m,2) - pow(n,2);
    b = 2*m*n;
    c = sqrt(pow(a,2)+pow(b,2));
    cout << a<< endl << b << endl << c << endl;
    return 0;
}
```

9- Write a C++ program that for a given number of apples, tells the user how many dozens of apples(s) he has and how many .extra apples are left over

```
# include <iostream>
using namespace std;

int main(){
    int apples, dozens;
    cout << "enter apples number: " , cin >> apples;
    dozens = apples / 12;
    apples = apples % 12;
    cout << dozens << " dozens and " << apples << " apples";
    return 0;
}
```

END OF SHEET 1

SHEET 2

1- Write a program that prints all solutions to the quadratic equation . (aX^2 + bX + C = 0) . Read in a, b, c and use the quadratic formula. If the discriminant is (b^2 – 4ac) negative, display a message stating that there are no solutions. Else solve it .

```
# include <iostream>
# include <cmath>
using namespace std;

int main(){
    float a, b, c, x, x1, x2;
    cout << "enter a value: " , cin >> a;
    cout << "enter b value: " , cin >> b;
    cout << "enter c value: " , cin >> c;
    switch((b*b) > (4*a*c)){
        case 1:
            x1 = (-b + sqrt((b*b)+4*a*c))/2*a;
            x2 = (-b + sqrt((b*b)-4*a*c))/2*a;
            cout << "x1: " << x1 << endl;
            cout << "x2: " << x2;break;
        default:cout << "equation not solvable";break;
    }
    return 0;
}
```

2- A year with 366 days is called a leap year. A year is a leap year if it is divisible by four (for example, 1980), except that it is not a leap year if it is divisible by 100 (for example, 1900); however, it is a leap year if it is divisible by 400 (for example, 2000). There were no exceptions before the introduction of the Gregorian calendar on October 15, 1582 (1500 was a leap year). Write a program that asks the user for a year and computes whether that year is a leap year. Your program should contain a single if statement.

```
# include <iostream>
using namespace std;

int main(){
    int x, y;
    cout << "enter year: " , cin >> x;
    if (((x>=1582 && (x%4 == 0 && x%100 != 0)) || (x%400 == 0))
|| (x < 1582 && x&4 ==0)){
        cout << "it's a leap year";
    }
    else{
        cout << "it's not a leap year";
    }
    return 0;
}
```

3-Write a program that asks the user to enter a month (1 for January, 2 for February, and so on) and then prints the number of days in the month. For February, print "28 or 29 days".

```
# include <iostream>
using namespace std;

int main() {
    int input;
    cin >> input;
    switch (input) {
    case 1:case 3:case 5:case 7:
    case 8:case 10:case 12:cout << "31 days.";break;
    case 4:case 6:case 11:cout << "30 days.";break;
    case 2:cout << "28 or 29 days.";break;
    }
    return 0;
}
```

4- :Assume we want to implement a football game. A player can do the following:

- ***Press "w" Move up***
- ***Press "s" Move down***
- ***Press "d" Move right***
- ***Press "a" Move left***
- ***Press "k" or "j" Kick the ball***
- ***Press "p" Pass the ball***
- ***Press "o" Over the ball***

This should be case insensitive so when user inputs "w" or "W", it should move up. This should apply for all entered letters; both upper and lower case characters .should work

---

```
# include <iostream>
# include <cctype>
using namespace std;

int main(){
    char move;
    cout << "choose your move: ", cin >> move;
    switch(tolower(move)){
        case 'w':cout << "Move up" ;break;
        case 's':cout << "Move down" ;break;
        case 'd':cout << "Move right" ;break;
        case 'a':cout << "Move left" ;break;
        case 'k':cout << "Kick the ball" ;break;
        case 'j':cout << "Kick the ball" ;break;
        case 'p':cout << "pass the ball" ;break;
        case 'o':cout << "Over the ball" ;break;
        default :cout << "Invalid input";
    }
    return 0;
}
```

5- Write a C++ program that designs a calculator .

The program should read from the user two integer numbers and a character which indicates the type of operation desired.

Hint : solve it by switch

```
# include <iostream>
using namespace std;

int main(){
    int x, y;
    char sign;
    cout << "enter number: ", cin >> x;
    cout << "enter sign: ", cin >> sign;
    cout << "enter number: ", cin >> y ;
    switch(sign){
        case '+':cout << "result: " << (x+y);break;
        case '-':cout << "result: " << x-y;break;
        case '*':cout << "result: " << x*y;break;
        case '/':cout << "result: " << x/y;break;
        default:cout << "Invalid input";
    }
    return 0;
}
```

***6- How can you find the minimum/maximum of three numbers using the conditional operator ,in just one line ?*

Hint : Google for ternary operator in C++

```
# include <iostream>
using namespace std;

int main(){
    int min,max;
    int n1,n2,n3;
    cout << "enter n1 value: " ,cin >> n1;
    cout << "enter n2 value: " ,cin >> n2;
    cout << "enter n3 value: " ,cin >> n3;
    max = (n1 > n2) ? ((n1 > n3) ? n1 : n3) : ((n2 > n3) ? n2 :
n3);
    min = (n1 < n2) ? ((n1 < n3) ? n1 : n3) : ((n2 < n3) ? n2 :
n3);
    cout << "max is: " << max << endl;
    cout << "min is: " << min;
    return 0;
}
```